

Project work part 1 - IND320

Malk Adwan

Imports and load data

```
import pandas as pd
import matplotlib.pyplot as plt

#reading the file
df_file = pd.read_csv(r"C:\Users\malki\OneDrive - Norwegian University
of Life Sciences\UNI5\IND320-1\Project\data\reservoirs.csv")
```

Describing the dataset and preprocessing

```
#general information about the data
#size of the dataset
# the first 5 rows of the dataset
#information about the columns of the dataset

print(df_file.shape)
print(df_file.head())
print(df_file.info())

(14877, 11)
      dato_Id  omrType  omrnrr  iso_aar  iso_uke  fyllingsgrad
kapasitet_TWh \
0  1995-09-03      EL      4    1995      35      0.944721
21.079208
1  2020-11-15      EL      1    2020      46      0.974361
6.003264
2  2011-08-28      EL      4    2011      34      0.786499
21.079208
3  1998-07-05      EL      3    1998      27      0.793969
8.920932
4  2011-03-27      EL      4    2011      12      0.300820
21.079208

      fylling_TWh  neste_Publiseringdato  fyllingsgrad_forrige_uke \
0      19.913979      0001-01-01T00:00:00      0.936888
1       5.849344      2020-11-25T13:00:00      0.997406
2      16.578772      0001-01-01T00:00:00      0.787193
3       7.082947      0001-01-01T00:00:00      0.733812
4       6.341054      0001-01-01T00:00:00      0.311122

      endring_fyllingsgrad
0          0.007833
1         -0.023046
2         -0.000694
3          0.060157
```

```

4          -0.010301
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 14877 entries, 0 to 14876
Data columns (total 11 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   dato_Id                               14877 non-null  object
1   omrType                               14877 non-null  object
2   omrn timer                             14877 non-null  int64
3   iso_aar                               14877 non-null  int64
4   iso_uke                               14877 non-null  int64
5   fyllingsgrad                         14877 non-null  float64
6   kapasitet_TWh                       14877 non-null  float64
7   fylling_TWh                         14877 non-null  float64
8   neste_Publiseringssdato             14877 non-null  object
9   fyllingsgrad_forrige_uke            14877 non-null  float64
10  endring_fyllingsgrad                 14877 non-null  float64
dtypes: float64(5), int64(3), object(3)
memory usage: 1.2+ MB
None

```

```

#describing the dataset in a more statistical sense
#along with number of observations for each column
print(df_file.describe())
print(df_file.isna().sum())

```

	omrnr	iso_aar	iso_uke	fyllingsgrad
count	14877.000000	14877.000000	14877.000000	14877.000000
mean	2.333333	2010.346038	26.405929	0.618852
std	1.490762	9.146696	15.017910	0.214850
min	0.000000	1995.000000	1.000000	0.055160
25%	1.000000	2002.000000	13.000000	0.456234
50%	2.000000	2010.000000	26.000000	0.648049
75%	3.000000	2018.000000	39.000000	0.802135
max	5.000000	2026.000000	53.000000	1.020072

	fylling_TWh	fyllingsgrad_forrige_uke	endring_fyllingsgrad
count	14877.000000	14877.000000	14877.000000
mean	18.139713	0.618890	-0.000038
std	15.966211	0.214843	0.031385
min	0.331142	0.055160	-0.242484

25%	7.321973		0.456356	-0.022395
50%	14.501389		0.648114	-0.007072
75%	22.203530		0.802133	0.016277
max	84.544820		1.020037	0.336623
dato_Id		0		
omrType		0		
omrnr		0		
iso_aar		0		
iso_uke		0		
fyllingsgrad		0		
kapasitet_TWh		0		
fylling_TWh		0		
neste_Publiseringsdato		0		
fyllingsgrad_forrige_uke		0		
endring_fyllingsgrad		0		
dtype: int64				

The dataset looks complete in the sense of no missing values in any of the columns. Stay tuned to find out if this truly is the case. The first registration was made in 1996 (min-value for iso-aar) while the latest registration was made in 2026.

Some of the headers explain themselves, others are not that easy to interpret.

```
df = df_file #renaming

#renaming the columns of the dataset
df = df.rename(columns={'dato_Id': 'Date', "omrType": "Area_type",
"omrnr": "Area_number",
                        "iso_aar": "ISO_year", "iso_uke": "ISO_week",
                        "fyllingsgrad": "Filling_level",
"kapasitet_TWh": "Capacity_TWh",
                        "fylling_TWh": "Filling_TWh",
                        "neste_Publiseringsdato": "Next_publishdate",
                        "fyllingsgrad_forrige_uke":
"Filling_level_prev_week",
                        "endring_fyllingsgrad":
"Change_filling_level"})

#sorting the dataframe by date so the datapoints become sequential
df = df.sort_values("Date")

#checking if the changes worked
df.head()
```

	Date	Area_type	Area_number	ISO_year	ISO_week
Filling_level \					
3468	1995-01-08	EL	5	1995	1
0.616448					
3361	1995-01-08	EL	4	1995	1
0.558864					

5036	1995-01-08	EL	3	1995	1
0.602376					
3113	1995-01-08	EL	1	1995	1
0.621467					
7739	1995-01-08	EL	2	1995	1
0.630947					

	Capacity_TWh	Filling_TWh	Next_publishdate
Filling_level_prev_week \			
3468	17.390587	10.720388	0001-01-01T00:00:00
0.614288			
3361	21.079208	11.780413	0001-01-01T00:00:00
0.801348			
5036	8.920932	5.373754	0001-01-01T00:00:00
0.596027			
3113	6.003264	3.730831	0001-01-01T00:00:00
0.734888			
7739	34.043590	21.479713	0001-01-01T00:00:00
0.777604			

	Change_filling_level
3468	0.002160
3361	-0.242484
5036	0.006349
3113	-0.113421
7739	-0.146657

We see that "Next_publishdate" for some measurements are 0001-01-01T00:00:00, which doesn't give any intuitive sense, but it could be that before that date, the system wasn't digitalised. This date is then the minimum date.

We can observe that the column "Date" is in the format of YYYY-MM-DD, but the type is an object, and the same is for "Next_Publishdate".

```
# converting the date columns to datetime format
for col in ["Date", "Next_publishdate"]:
    df[col] = pd.to_datetime(df[col], format="ISO8601",
errors="coerce")

print(df[["Date", "Next_publishdate"]].dtypes)

Date          datetime64[ns]
Next_publishdate  datetime64[ns]
dtype: object

#checking if the changes were made
df.head()
```

	Date	Area_type	Area_number	ISO_year	ISO_week
Filling_level \					

3468	1995-01-08	EL	5	1995	1
0.616448					
3361	1995-01-08	EL	4	1995	1
0.558864					
5036	1995-01-08	EL	3	1995	1
0.602376					
3113	1995-01-08	EL	1	1995	1
0.621467					
7739	1995-01-08	EL	2	1995	1
0.630947					

	Capacity_TWh	Filling_TWh	Next_publishdate
Filling_level_prev_week \			
3468	17.390587	10.720388	NaT
0.614288			
3361	21.079208	11.780413	NaT
0.801348			
5036	8.920932	5.373754	NaT
0.596027			
3113	6.003264	3.730831	NaT
0.734888			
7739	34.043590	21.479713	NaT
0.777604			

	Change_filling_level
3468	0.002160
3361	-0.242484
5036	0.006349
3113	-0.113421
7739	-0.146657

NaT = Not a Time. Happens when there are missing dates within a dataframe. 0001-01-01T00:00:00 is not a date and the value therefore becomes NaT. So even though `isna().sum()` gave us that there is no missing values, by actually inspecting the dataset we see that there is indeed missing values.

Plotting each column

```
print(df["Area_type"].unique()) #checking unique values in the
Area_type column
print(df.groupby("Area_type")["Area_number"].unique())

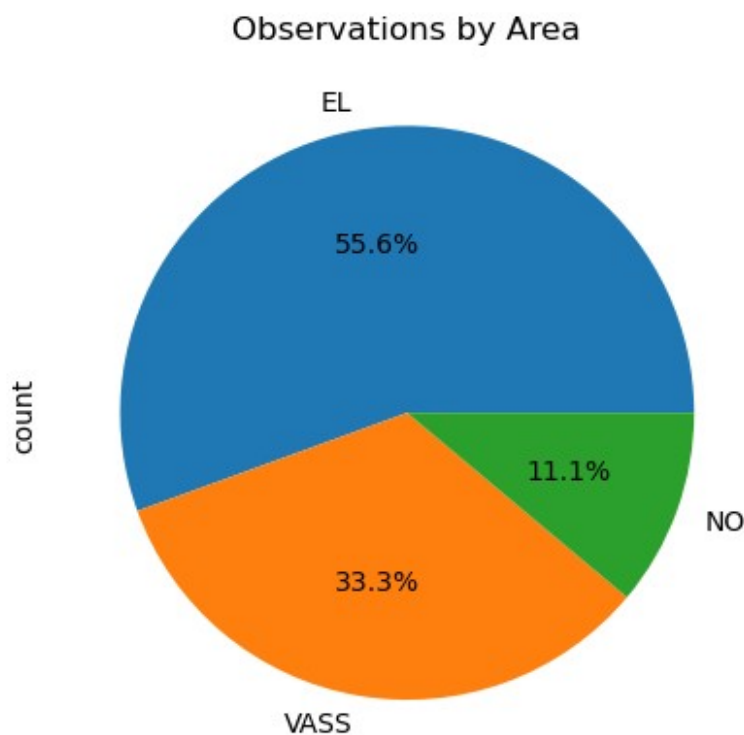
['EL' 'VASS' 'NO']
Area_type
EL      [5, 4, 3, 1, 2]
NO      [0]
VASS    [2, 1, 3]
Name: Area_number, dtype: object
```

This columns seems to hold information about how the data is categorized in areas, wether it is

- NO: The whole country of Norway
- EL: one of the five electricity price areas
 - EL1
 - EL2
 - EL3
 - EL4
 - EL5
- VASS: river basin area (Vassdragsområde)
 - VASS1
 - VASS2
 - VASS3

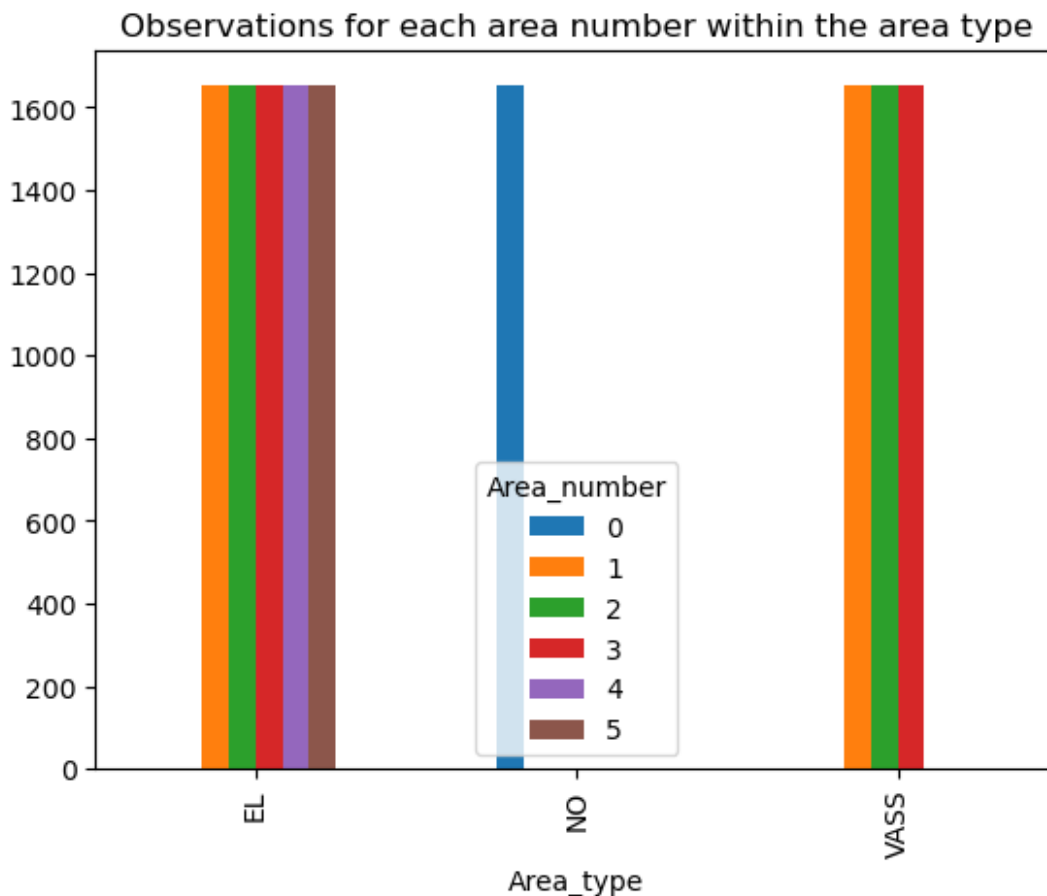
```
# how many observations are made for each area type
df["Area_type"].value_counts().plot(kind="pie", autopct="%1.1f%%")
plt.title("Observations by Area")
```

```
Text(0.5, 1.0, 'Observations by Area')
```



```
#different area numbers for each area type
df.groupby(["Area_type",
"Area_number"]).size().unstack(fill_value=0).plot(kind="bar")
plt.title("Observations for each area number within the area type")
```

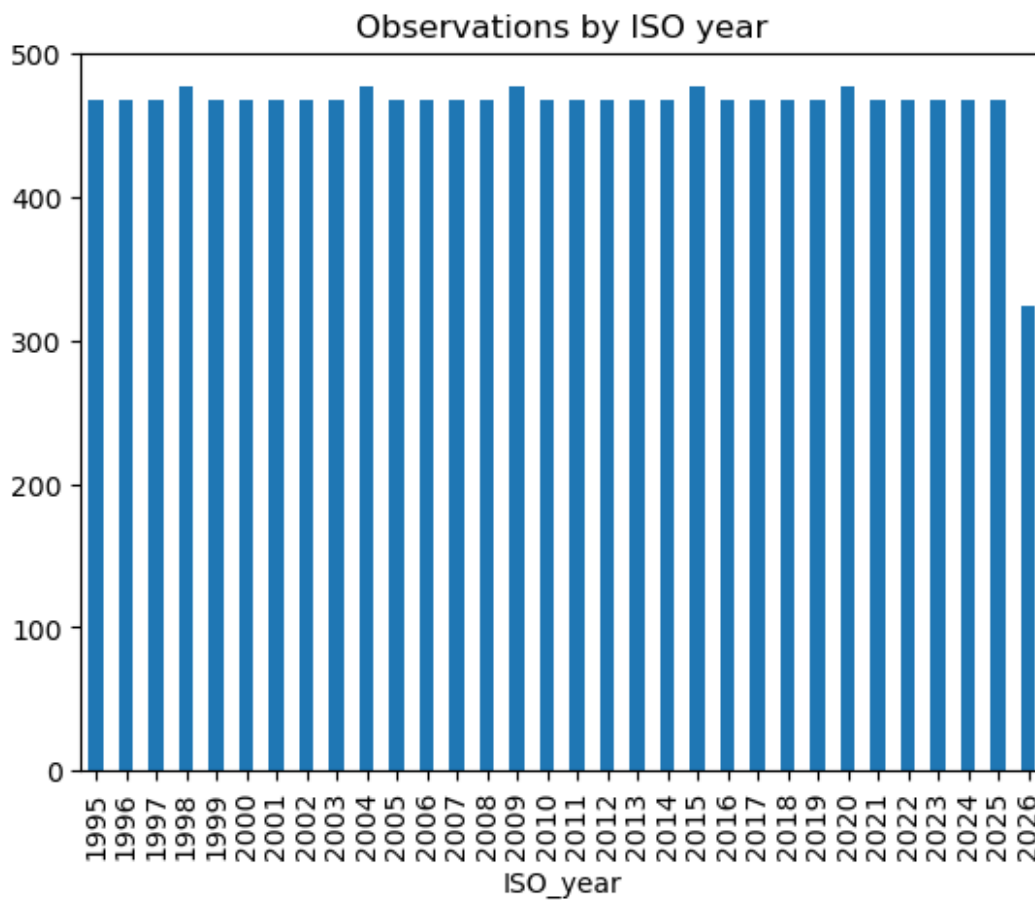
```
Text(0.5, 1.0, 'Observations for each area number within the area type')
```



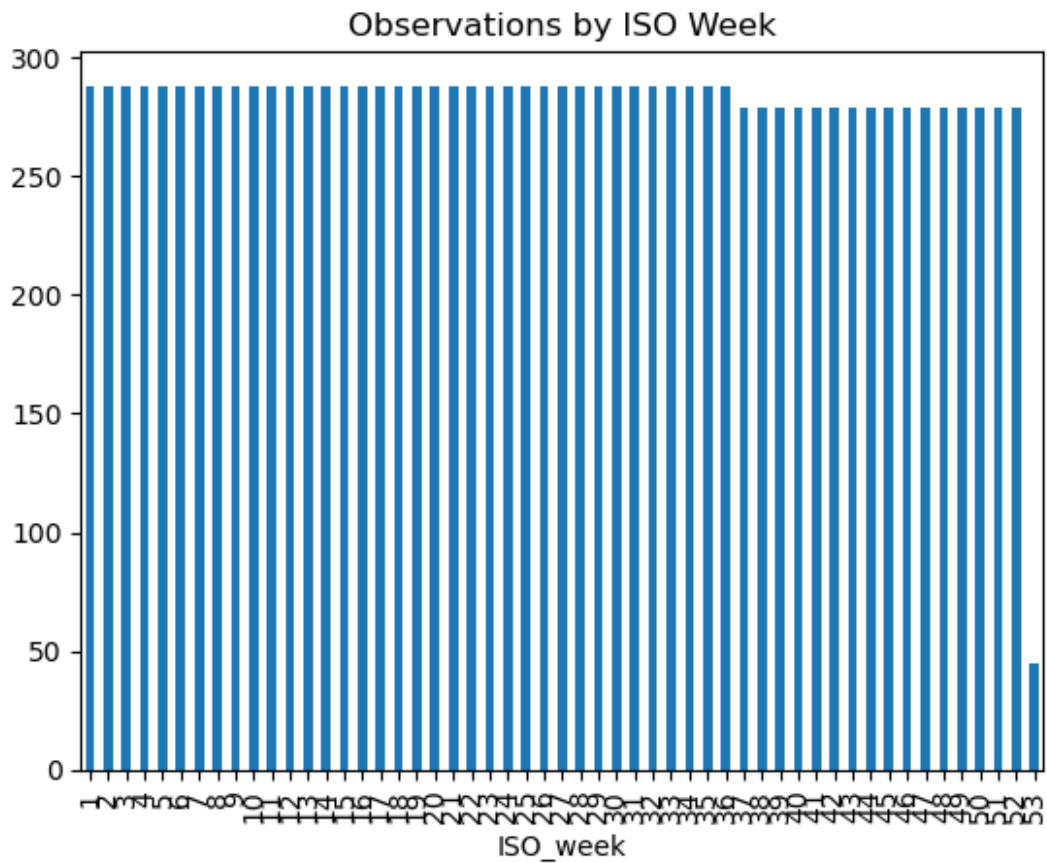
NO has no area number as it covers the whole country, so the default area number is zero. However, each area number is represented in the same amount, this can also be seen in the pie chart.

```
#how many observations there are for each ISO year
#ISO years are essentially observations throughout the years, grouped by year
df["ISO_year"].value_counts().sort_index().plot(kind="bar")
plt.title("Observations by ISO year")

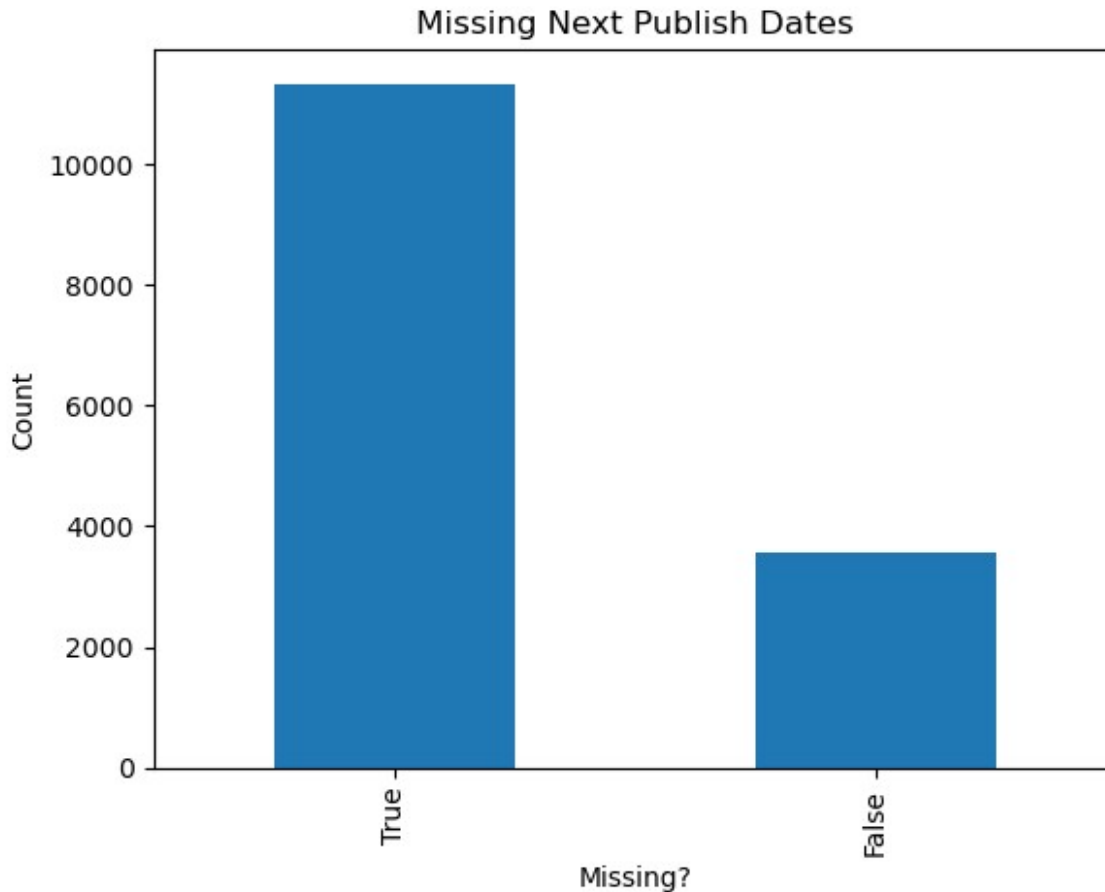
Text(0.5, 1.0, 'Observations by ISO year')
```



```
#how many observations there are for each ISO week  
#ISO weeks are essentially observations throughout the years, grouped  
by week number  
df["ISO_week"].value_counts().sort_index().plot(kind="bar")  
plt.title("Observations by ISO Week")  
plt.show()
```



```
#how many of the Next_publishdate is NaT
df["Next_publishdate"].isna().value_counts().plot(kind="bar")
plt.title("Missing Next Publish Dates")
plt.xlabel("Missing?")
plt.ylabel("Count")
plt.show()
```



Even though we showed above that there was no missing values, looking further into the data, we now see that the missing values were replaced by a default date, and then when we transformed the column to have a proper date-format, the missing values became marked as NaT.

It doesn't make sense to plot the Date-column by itself, so instead I will group the measurements by month, and then look at when the measurements are taken throughout the seasons of the year.

```
#count the number of measurement for each month of the year  
#then assign the months an index from 1-12  
monthly_measurements = df["Date"].dt.month.value_counts().sort_index()  
  
#assign the indexed months to their respective season  
#sum all the observations for each month within the season  
spring_months = monthly_measurements[[3, 4, 5]].sum()  
summer_months = monthly_measurements[[6, 7, 8]].sum()  
fall_months = monthly_measurements[[9, 10, 11]].sum()  
winter_months = monthly_measurements[[12, 1, 2]].sum()  
  
#create a pandas series where each season get its corresponding  
#number of measurements
```

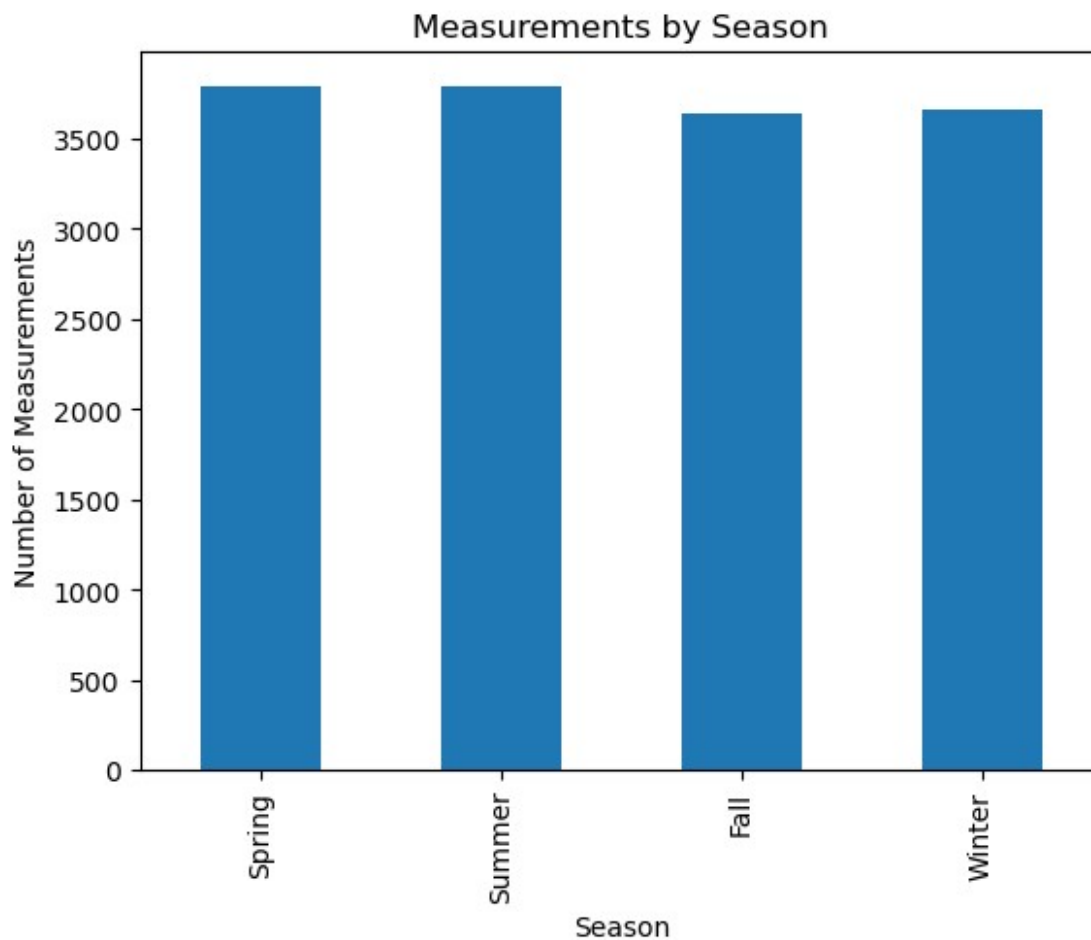
```

seasonal_measurements = pd.Series(
    [spring_months, summer_months, fall_months, winter_months],
    index=["Spring", "Summer", "Fall", "Winter"]
)

seasonal_measurements.plot(kind="bar")

plt.title("Measurements by Season")
plt.xlabel("Season")
plt.ylabel("Number of Measurements")
Text(0, 0.5, 'Number of Measurements')

```



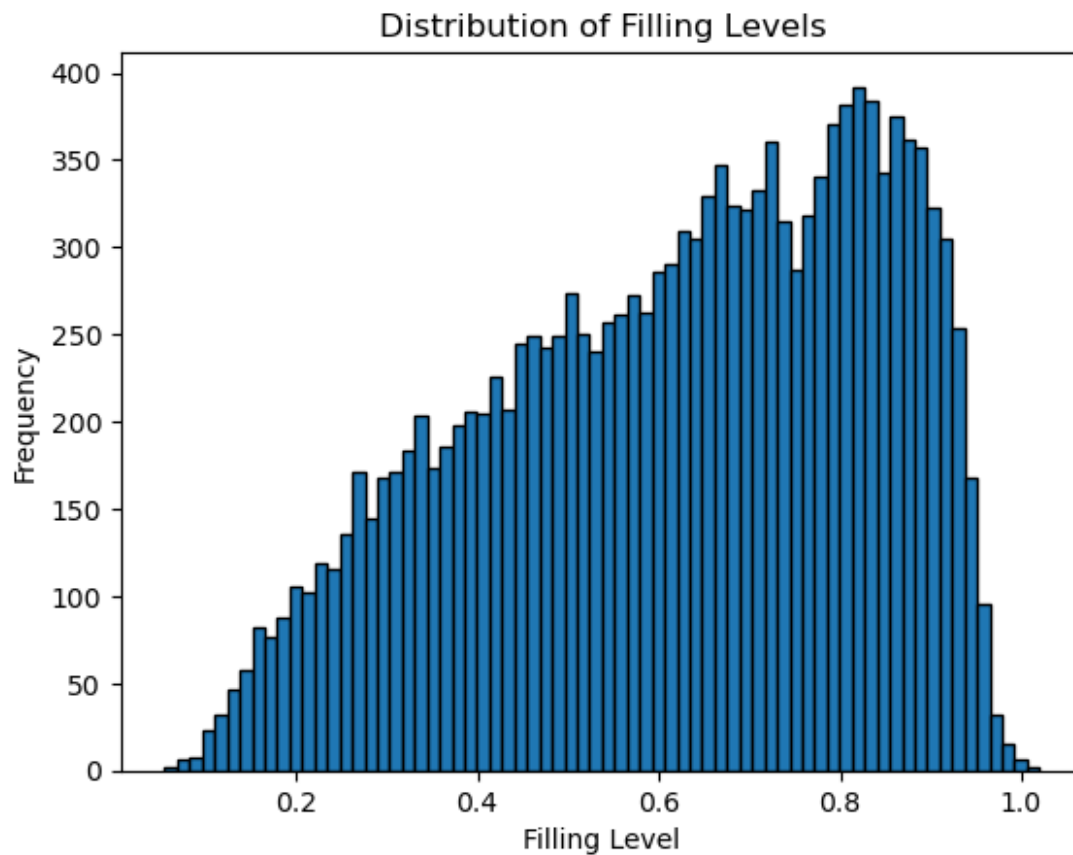
Judging by this, it looks like there are fewer measurements during the colder months in contrary to the summer and spring months. However, the difference doesnt seem significant.

```

df["Filling_level"].plot(kind="hist", bins=70, edgecolor="black")
#tried different bin sizes, 70 seems good
plt.title("Distribution of Filling Levels")
plt.xlabel("Filling Level")

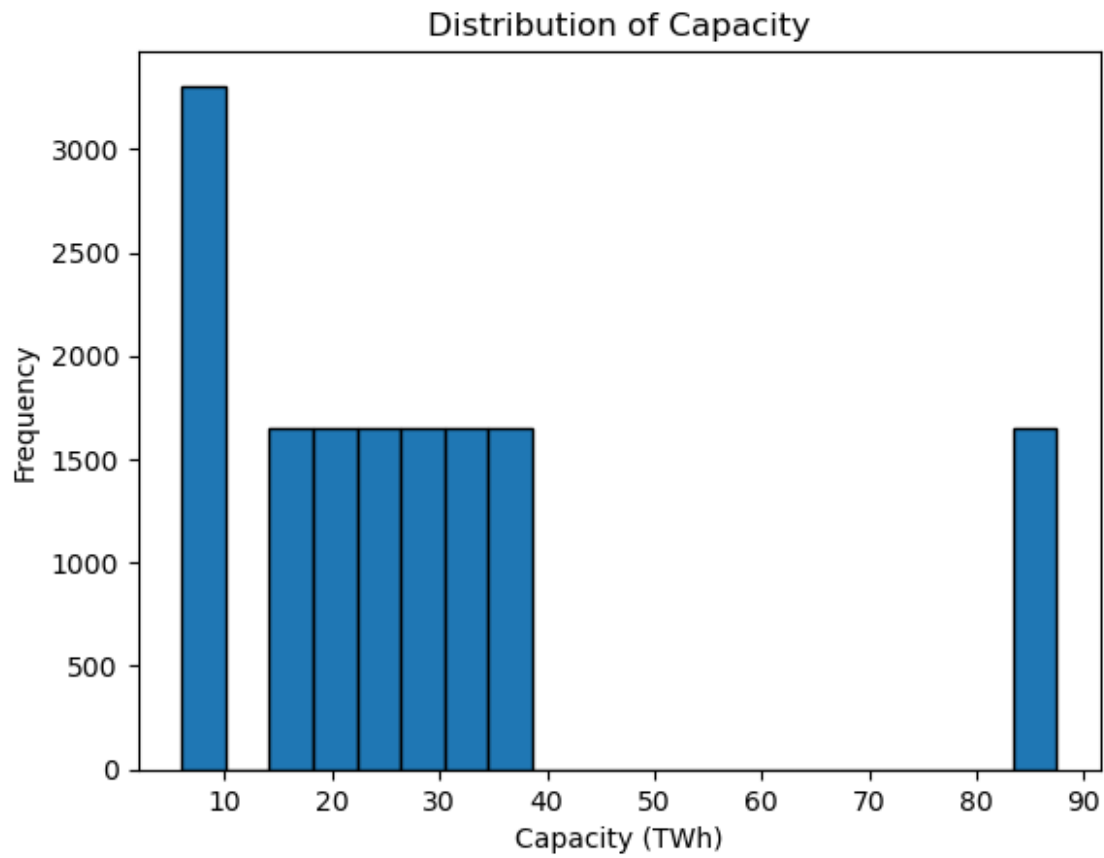
```

```
Text(0.5, 0, 'Filling Level')
```



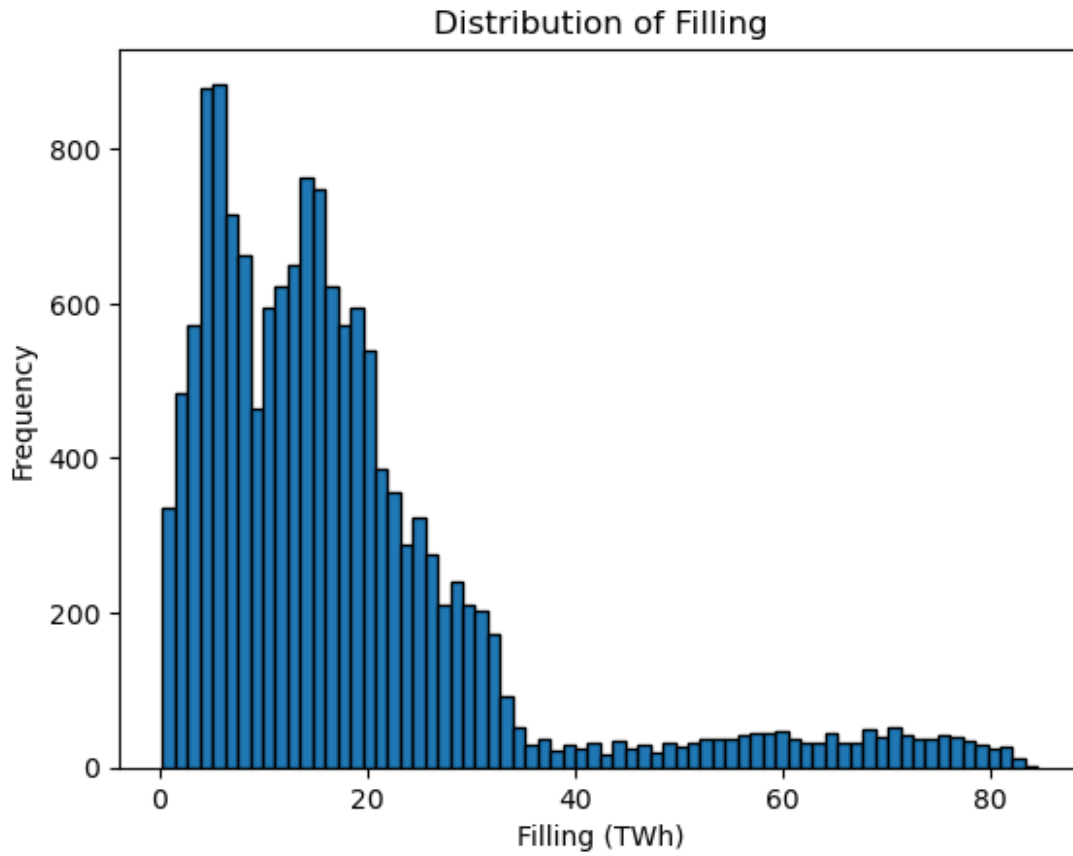
The distribution of the filling level is slightly left-skewed. This means that most reservoirs have relatively high filling levels, but there are some observations with unusually low filling levels.

```
df["Capacity_TWh"].plot(kind="hist", bins=20, edgecolor="black")  
  
plt.title("Distribution of Capacity")  
plt.xlabel("Capacity (TWh)")  
  
Text(0.5, 0, 'Capacity (TWh)')
```



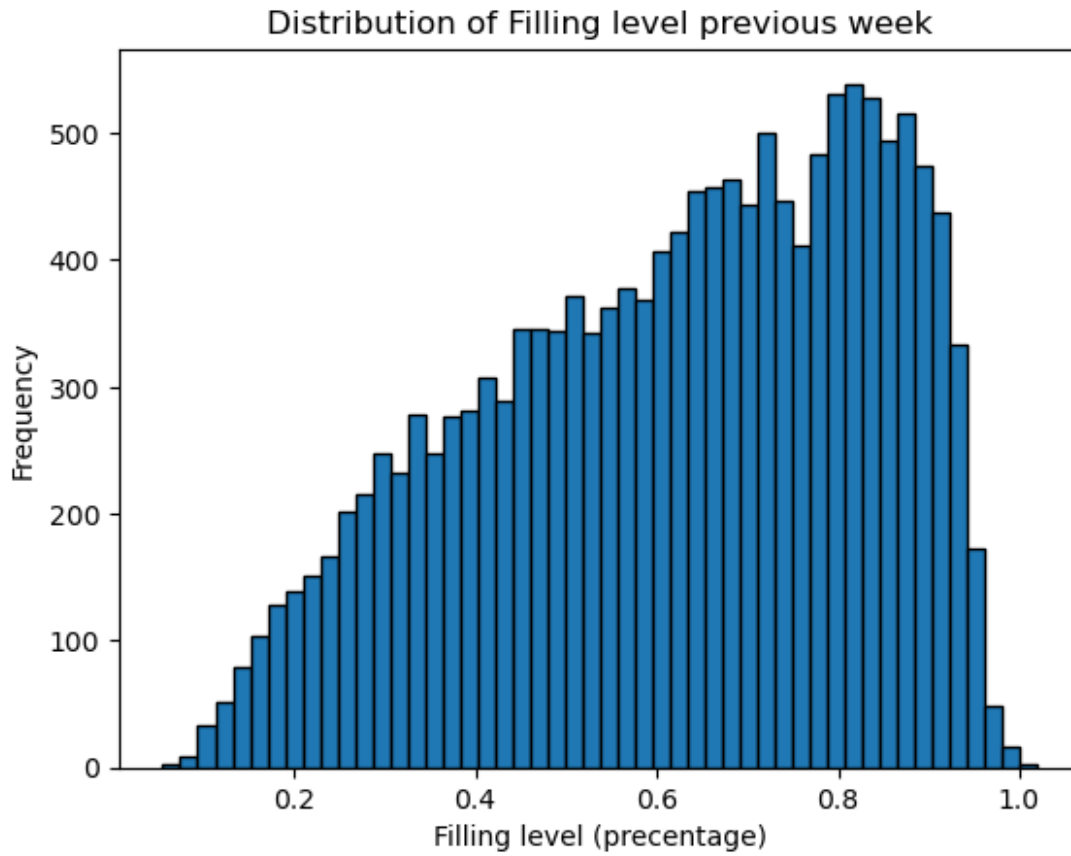
The distribution of capacity shows that most observations have a capacity of around 10 TWh. The remaining observations are mainly distributed between 15 and 40 TWh, with a few observations having a capacity of approximately 85 TWh.

```
df["Filling_TWh"].plot(kind="hist", bins=70, edgecolor="black")  
  
plt.title("Distribution of Filling")  
plt.xlabel("Filling (TWh)")  
  
Text(0.5, 0, 'Filling (TWh)')
```



The distribution of Filling TWh is right-skewed. This means that most observations are concentrated at lower values, while a smaller number of observations have higher values, creating a longer tail toward the right.

```
df["Filling_level_prev_week"].plot(kind="hist", bins=50,  
edgecolor="black")  
  
plt.title("Distribution of Filling level previous week")  
plt.xlabel("Filling level (percentage)")  
  
Text(0.5, 0, 'Filling level (percentage)')
```

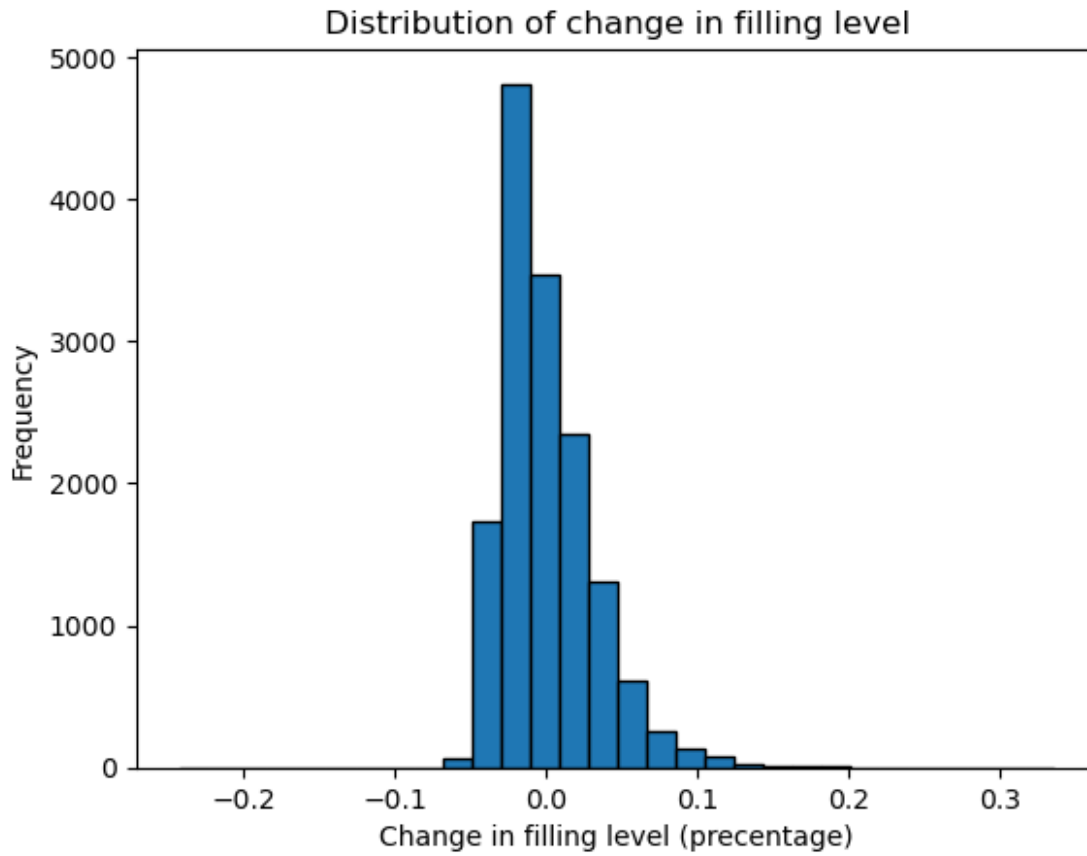


The distribution of the filling level for the previous week is slightly left-skewed. Most observations are concentrated toward the higher filling levels, while the observations gradually decrease toward the lower values. However, the skewness is not very pronounced.

```
df["Change_filling_level"].plot(kind="hist", bins=30,
edgecolor="black")

plt.title("Distribution of change in filling level")
plt.xlabel("Change in filling level (percentage)")

Text(0.5, 0, 'Change in filling level (percentage)')
```



```
# counting the number of observations for the change in filling level
below_zero = (df["Change_filling_level"] < 0).sum() #the change is
negative = less filling than last measured
above_zero = (df["Change_filling_level"] > 0).sum() #the change is
positive = more filling than last measured
at_zero = (df["Change_filling_level"] == 0).sum() #the change is zero
= no change from last time
```

```
print("Below 0:", below_zero)
print("Above 0:", above_zero)
print("Exactly 0:", at_zero)
```

```
Below 0: 8704
Above 0: 6173
Exactly 0: 0
```

Then we have the distribution of the change in filling level. Most observations are below zero, meaning on the left side, giving a distribution with a tail on right-side, making it right-skewed.

Plotting columns against each other

Plotting the measurements over time, where the datapoints are grouped by area_type and then looking at area_number for each area_type.

```

el_area = df[df["Area_type"] == "EL"].copy() #filtering the dataframe
to only include rows where Area_type is "EL"
no_area = df[df["Area_type"] == "NO"].copy() #filtering the dataframe
to only include rows where Area_type is "NO"
vass_area = df[df["Area_type"] == "VASS"].copy() #filtering the
dataframe to only include rows where Area_type is "VASS"

#create 3 plots, one for each Area_type
#the plots are stacked vertically (3 rows, 1 column)
fig, axes = plt.subplots(3, 1, figsize=(12, 10),
    sharex=True, #use the same Date axis for all plots
    sharey=True #use the same Filling level scale for all plots
)

#loop through each Area_type and its corresponding plot
for ax, area_type in zip(axes, df["Area_type"].unique()):

    #select only observations belonging to the current Area_type
    #and sort them chronologically by Date
    subset = df[df["Area_type"] == area_type].sort_values("Date")

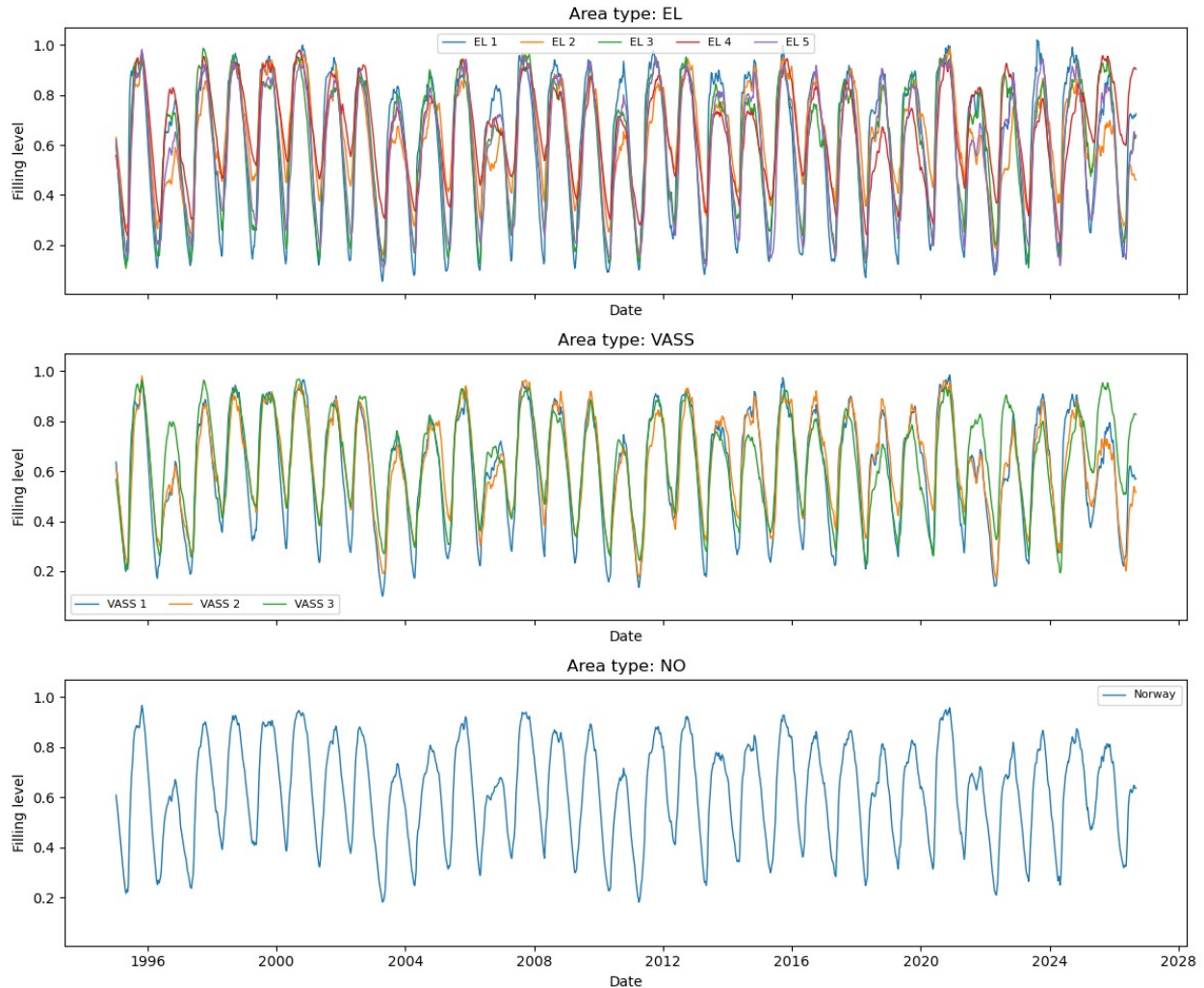
    #group the data by Area_number
    for number, area in subset.groupby("Area_number"):
        #create a label for the legend
        #if the area type is NO, call it "Norway"
        #else use the Area_type and Area_number
        label = "Norway" if area_type == "NO" else f"{area_type}
{number}"

        #plot the Filling_level over Date for the current area
        ax.plot(
            area["Date"],
            area["Filling_level"],
            label=label,
            linewidth=1
        )

    #Adding a title showing which Area_type is being displayed
    ax.set_title(f"Area type: {area_type}")
    ax.set_ylabel("Filling level")
    ax.set_xlabel("Date")
    #legend showing which line belongs to which area
    ax.legend(ncol=5, fontsize=8)

plt.tight_layout()
plt.show()

```



```
#Lisy of measurement columns that we want to plot
measure_cols = ["Filling_TWh", "Filling_level_prev_week",
                "Change_filling_level", "Capacity_TWh"]

#Looping through the columns
#for each measure, a figure is created
for col in measure_cols:
    #3 subplots are created for each area type
    fig, axes = plt.subplots(3, 1, figsize=(12, 9), sharex=True)

    #loop through the area tyoes and their df
    for ax, area_df, area_type in zip(axes, [no_area, el_area,
vass_area], ["NO", "EL", "VASS"]):
        #sirting the data by date and then group by Are_number
        for number, g in
area_df.sort_values("Date").groupby("Area_number"):
            #plot the selected measurement over time
            #each area number gets its line
```

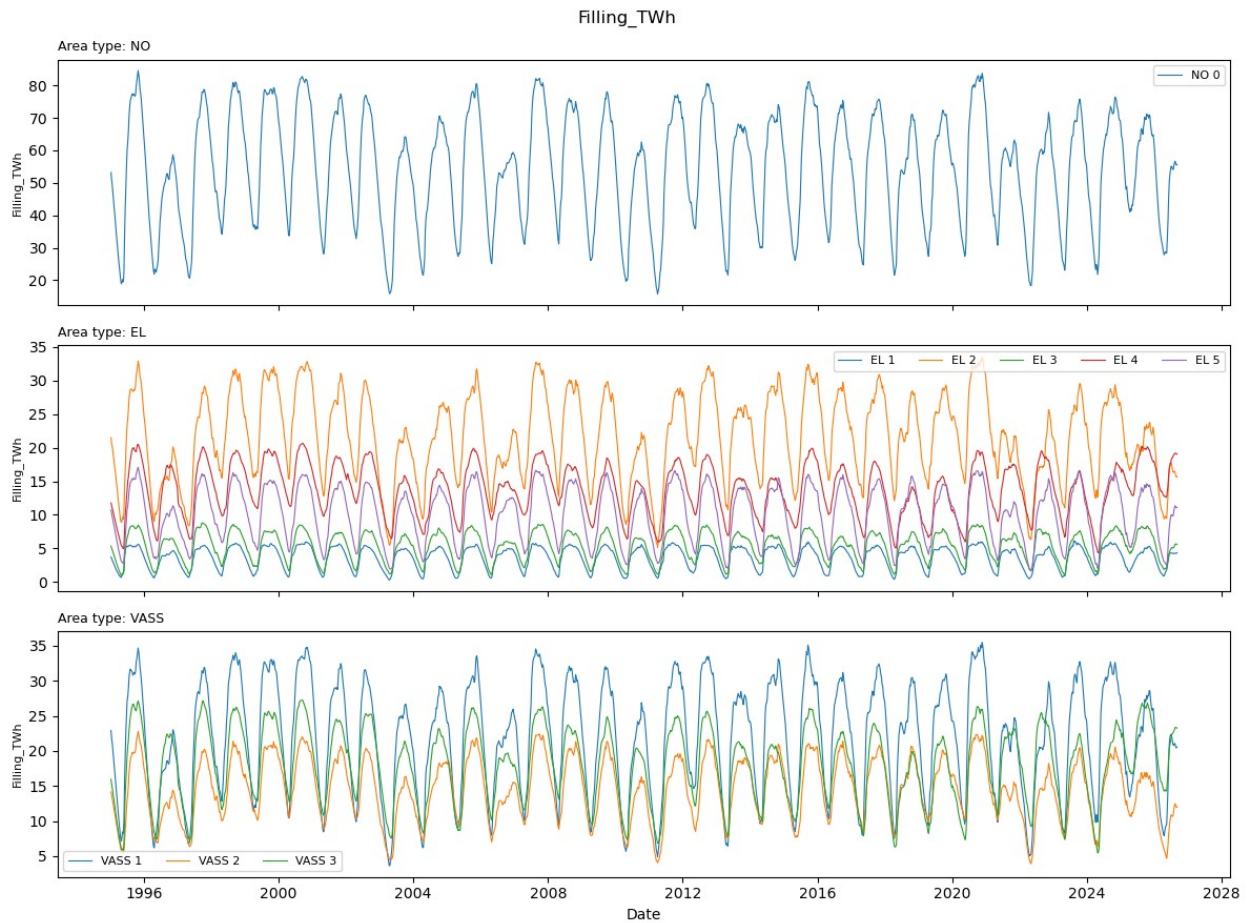
```

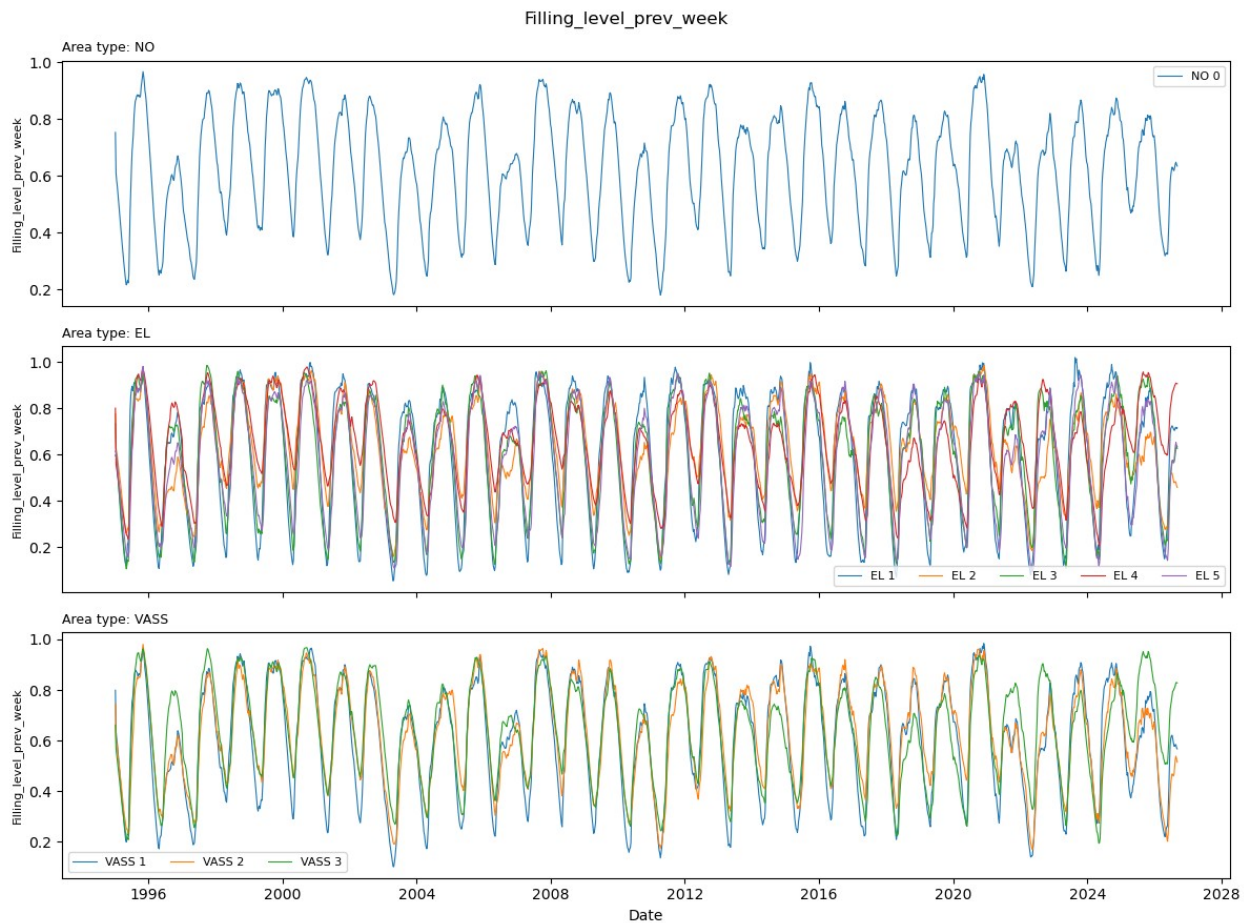
ax.plot(g["Date"], g[col], label=f"{area_type} {number}",
linewidth=0.8)

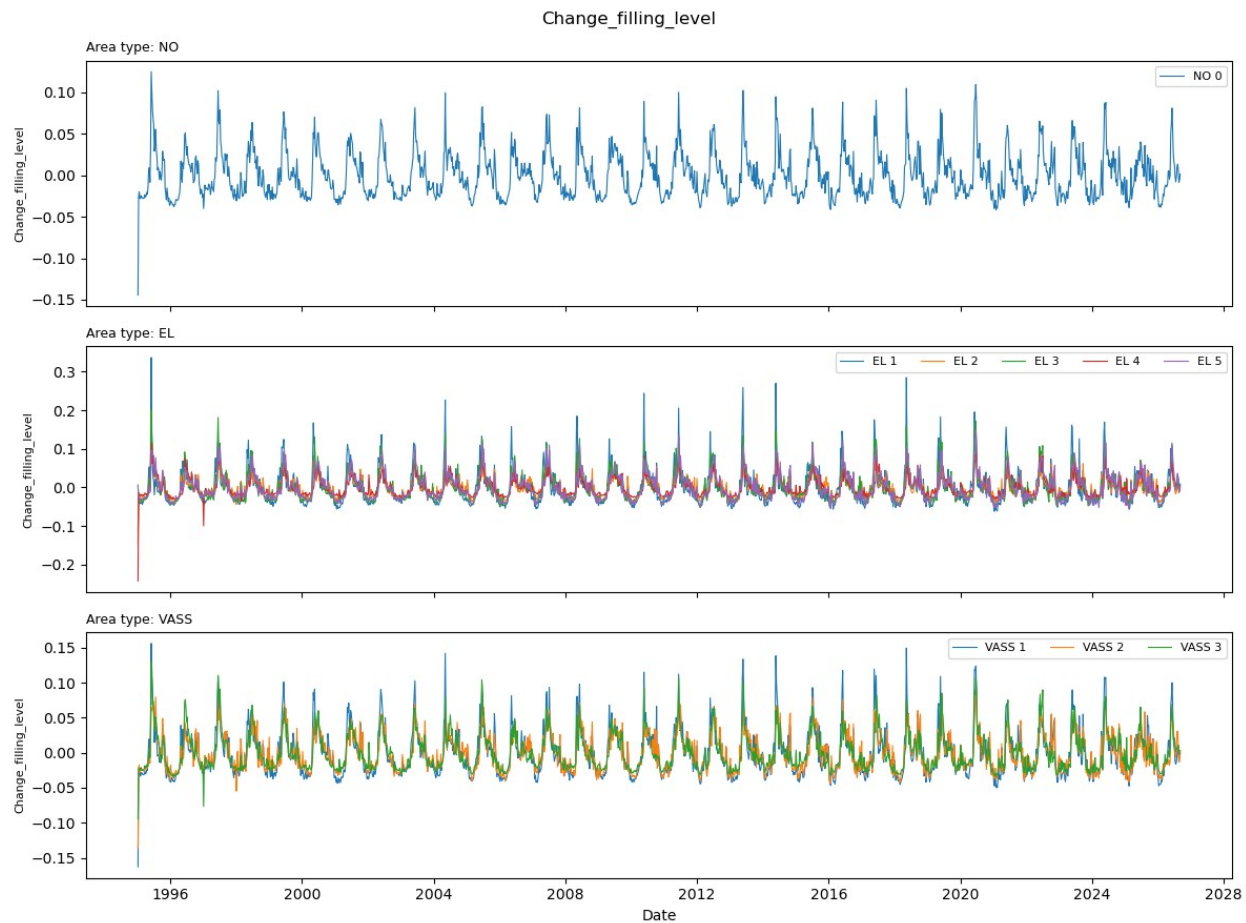
#which Area_type is being shown
ax.set_title(f"Area type: {area_type}", loc="left",
fontsize=9)
ax.set_ylabel(col, fontsize=8)
ax.legend(ncol=5, fontsize=8)

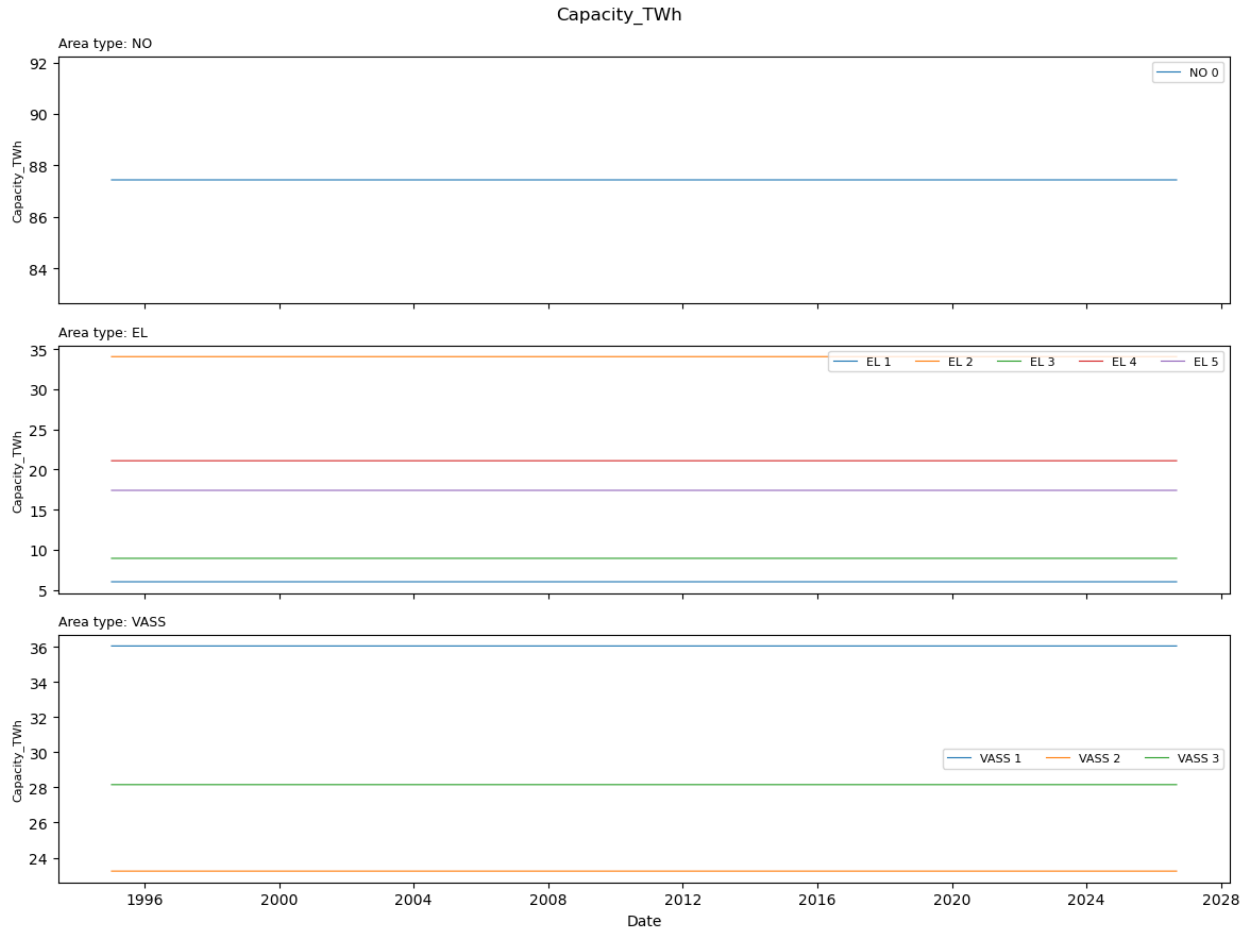
axes[-1].set_xlabel("Date") #naming the bottom x-axis
#naming the figure the selected measurement
fig.suptitle(col)
plt.tight_layout()
plt.show()

```









- Filling (TWh) against date: The different colors represent the different area numbers within each area type. VASS-1, EL-2 and NO appear to follow similar trends over time, although they are on different scales. This suggests that the filling levels in these areas tend to increase and decrease in similar periods.
- Filling level in the previous week against date: Here, EL-5, VASS-2 and NO appear to follow similar trends over time. The filling level in the previous week also shows relatively consistent patterns between the different areas.
- Change in filling level against date: The changes in filling level appear to be relatively small throughout the observed period. There are some larger peaks, particularly around 1996. These peaks could indicate periods with larger changes in reservoir filling levels.
- Capacity (TWh) against date: The capacity appears to remain constant for the different area types and area numbers throughout the observed period. This is expected because the capacity represents the maximum storage capacity of the reservoirs rather than the amount of water currently stored. The graphs also show that the same area-number series are present throughout the period.

Plotting all the columns for one area_type

Only plotting all the columns for Area_type="NO" so it doesn't look as chaotic.

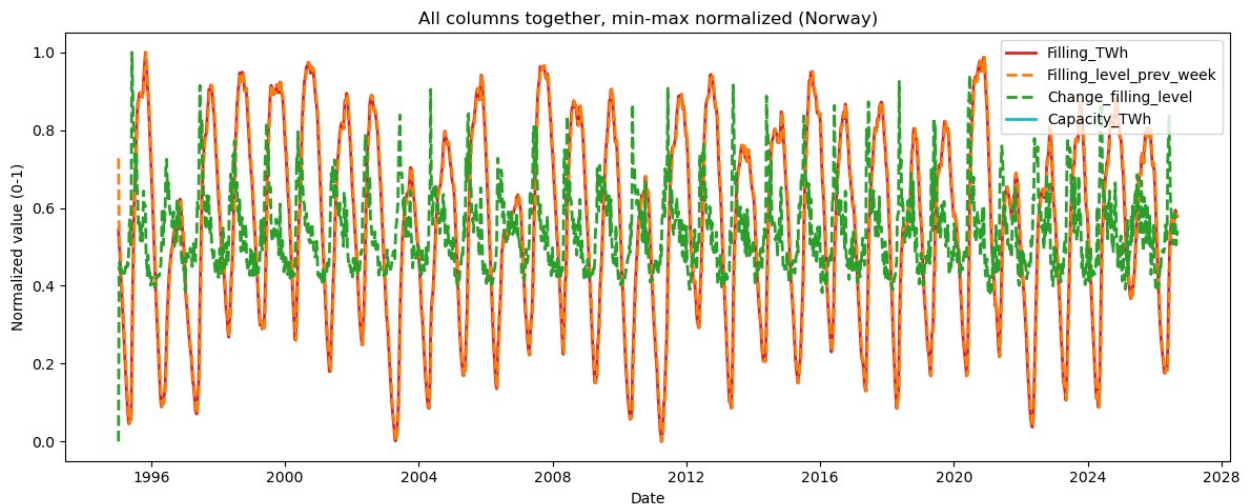
```
#normalizes all the measurement column to a scale from 0-1
#Makes the columns comparable since they have different scales.
norm = (no_area[measure_cols] - no_area[measure_cols].min()) /
(no_area[measure_cols].max() - no_area[measure_cols].min())

#different line styles to easier differentiate the lines
styles = ["-", "--", "-.-", "-"]
#different colors for the lines
colors = ["tab:red", "tab:orange", "tab:green", "tab:cyan"]

fig, ax = plt.subplots(figsize=(12, 5))

#looping through each measurement column, linestyle and color
for col, style, color in zip(measure_cols, styles, colors):
    #Plotting the normalized measurements over time
    ax.plot(no_area["Date"], norm[col], label=col, linewidth=2.0,
            linestyle=style, color=color)

ax.set_xlabel("Date")
ax.set_ylabel("Normalized value (0-1)")
ax.set_title("All columns together, min-max normalized (Norway)")
ax.legend()
plt.tight_layout()
plt.show()
```



```
#calculate the correlation between the normalized measurements
correlations = norm.corr()

print(correlations)
```

	Filling_TWh	Filling_level_prev_week	\
Filling_TWh	1.000000	0.990009	
Filling_level_prev_week	0.990009	1.000000	
Change_filling_level	0.069798	-0.071562	
Capacity_TWh	NaN	NaN	

	Change_filling_level	Capacity_TWh
Filling_TWh	0.069798	NaN
Filling_level_prev_week	-0.071562	NaN
Change_filling_level	1.000000	NaN
Capacity_TWh	NaN	NaN


```

#describing the Capacity column for area type NO
#since we get NaN-values
print(no_area["Capacity_TWh"].describe())

```

```

count    1.653000e+03
mean      8.743758e+01
std       1.378870e-12
min       8.743758e+01
25%      8.743758e+01
50%      8.743758e+01
75%      8.743758e+01
max       8.743758e+01
Name: Capacity_TWh, dtype: float64

```

The normalized measurements show that Filling TWh and Filling level in the previous week follow very similar patterns over time. This is also supported by their very high correlation of 0.990. Change in filling level behaves differently and has very little relationship with the other two variables.

Capacity TWh is not visible in the normalized plot because its value is constant in the data for the area we examined. Since the capacity does not change, it cannot be normalized in the same way as the other variables. Therefore, we cannot use this plot to compare the development of capacity with the other measurements.

Overall, the results show that Filling TWh and Filling level in the previous week change in a very similar way, while Change in filling level follows a different pattern.

Log

I started by reading reservoirs.csv with pandas and exploring its structure. The original column headers were in Norwegian, so I renamed them to English equivalents using a rename dictionary, which made the rest of the analysis easier to write and read.

Working with the Date column surfaced an early issue: many rows contained the placeholder value 0001-01-01T00:00:00, which represents "no date set" but falls outside pandas' default datetime range. I handled this by converting the values to NaT instead of raising an error.

A more significant discovery was that Area_type splits the data into three region types — NO (national total), EL (five electricity price areas), and VASS (three watercourse areas) — and these

overlap rather than being independent. Plotting a column without filtering to one area mixed unrelated regions into a single, meaningless line. This meant every plot needed to group or filter by Area_type and Area_number before drawing, which shaped how I approached both the "plot each column" and "plot all columns together" requirements. For the latter, since the columns have very different scales, I applied min-max normalization so they could share one axis meaningfully.

For the Streamlit app, I set up a GitHub repository and connected it to Streamlit Community Cloud, which took some troubleshooting. Streamlit's GitHub App needed explicit permission to see my repository even after making it public, which I resolved by reconfiguring its access under GitHub's installed app settings. I also ran into a version mismatch between the pandas installed in my Jupyter kernel versus the environment streamlit run used locally, which caused format="ISO8601" to silently fail to parse dates, switching to unformatted pd.to_datetime resolved it across environments.

I built the four-page structure with a home page, a data table page using st.column_config.LineChartColumn() to show row-wise sparklines of the first month's data, and a plot page with st.selectbox for column choice and st.select_slider for a month range. I used @st.cache_data on the data-loading function so the CSV is only read once per session rather than on every rerun.

However, I feel that I may have misunderstood parts of the project, as the visualization in my ipynb was not the same as what I understood to be required in the Streamlit app.

AI usage

I used AI (Anthropic, 2026) assistance throughout for debugging error tracebacks (pandas dtype issues, a matplotlib import error, Streamlit-GitHub permission problems), for reasoning through data-structure questions (what the area types represent and why unfiltered plots looked wrong), and for scaffolding page code, which I then adapted to match my own notebook logic. I also used AI to formulate descriptions and the project log, as well as for language editing (språkvask) to improve wording, grammar, and clarity.

Anthropic. (2026). Claude (Sonnet 5) [Large language model]. <https://claude.ai>

GitHub and Streamlit app

GitHub repository: [malkia777/IND320---Project](https://github.com/malkia777/IND320---Project)

Streamlit app: <https://share.streamlit.io/user/malkia777>

Panopto link

<https://nmbu.cloud.panopto.eu/Panopto/Pages/Viewer.aspx?id=bb44b5d6-be35-4a17-9bb2-b4d4015e0fa7>